

Vacant

Executable Acceptance Testing for LLM Code Agent Delivery
Mechanism Comparisons Replication Experiments and Accountability Records in Vacant

2026 9 13 2026 9 14

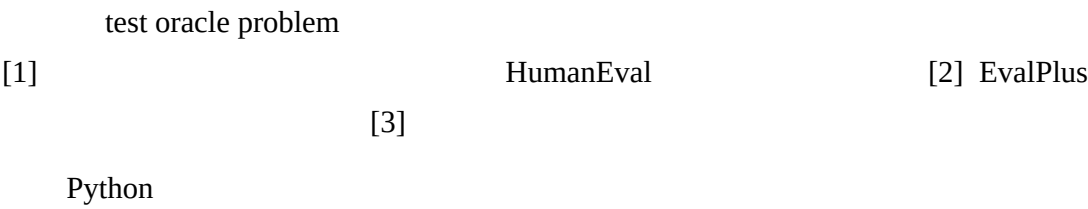
Vacant									
12B									
					Holm		14.17	19.17	
H-MIX	+5.83	+4.17	+0.83	+2.50	+4.31			Holm	716
H-MIX	+1.12	p = .3409			+7.96	p = 2.02 × 10 ⁻⁸			
194	9,841								

Abstract

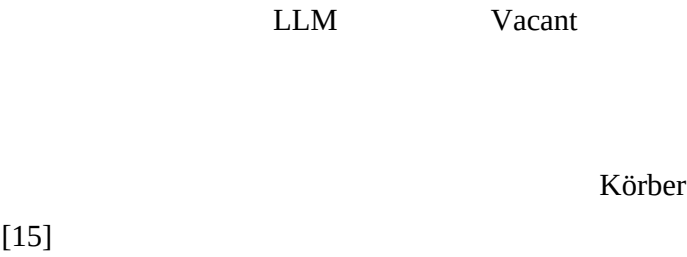
LLM code agents can generate programs, but their own claims of correctness do not establish a sound basis for delivery. We investigate how additional sampling, executable selection, and feedback-driven revision contribute to successful delivery when requirements are already represented by executable acceptance tests. Vacant integrates a visible-test delivery gate with hidden-test evaluation and signed hash-chain records. Using one locally hosted, quantized 12B model, we compare single-shot generation, gated resampling with up to five calls, five-candidate majority selection, and three revision workflows. The evaluation includes five new-seed repetitions on the same 120-task bank and a separate four-set evaluation comprising 716 tasks. Across the five repetitions, all fifteen revision-versus-single-shot comparisons pass within-repetition Holm correction. Gated resampling improves over single-shot generation by 14.17 to 19.17 percentage points. However, the primary H-MIX workflow improves over gated resampling by only 5.83, 4.17, 0.83, 2.50, and 4.31 percentage points; none of these five comparisons passes Holm correction. Across benchmark sets, the pooled improvement is 1.12 percentage points over gated resampling (adjusted p = .3409) and 7.96 points over single-shot generation (adjusted p = 2.02 × 10⁻⁸). All 9,841 entries in 194 archived receipt chains pass signature verification. These findings support executable gating with resampling under the evaluated conditions, while the incremental benefit of revision

remains unresolved. The contribution is an auditable integration and empirical comparison, rather than a new repair algorithm or a general guarantee of correctness. Limited test coverage, repeated tasks, a single model, and heterogeneous inference backends constrain generalization.

1.1 □□□□□□□



1.2 □□□□□□□



1.3 □□□□□□□

2.1 □□□□□□□□□



MBPP+ HumanEval+ LiveCodeBench
LCB v2 v3

Barr Vacant [1]

2.2 ☐☐☐☐☐☐☐☐☐☐

Self-consistency [5] OFF5

chain-of-thought

Self-Debugging [6] Reflexion [7]

H-MIX

Olausson [8] HumanEval APPS

coding-agent

2.3 ☐☐☐☐☐☐☐☐☐☐

Kim 350 [9]

DRF

[10] Ebrahimi

[11] He A-Trust

[12]

Vacant

Vacant

2.4 ☐☐☐☐☐☐☐☐☐☐

PeerReview [13] Vacant

PeerReview

Certificate Transparency [14] Vacant

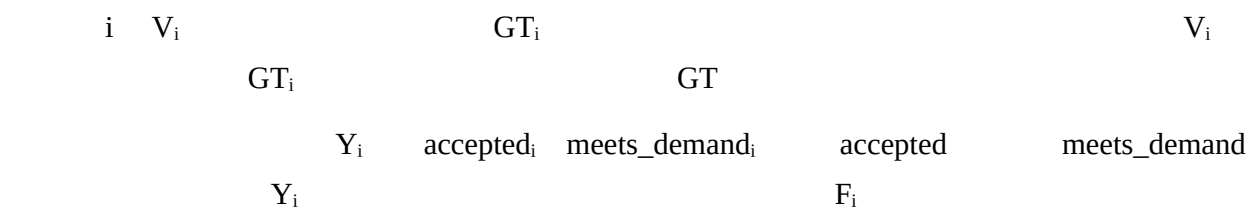
CT Merkle

Nosek [16]

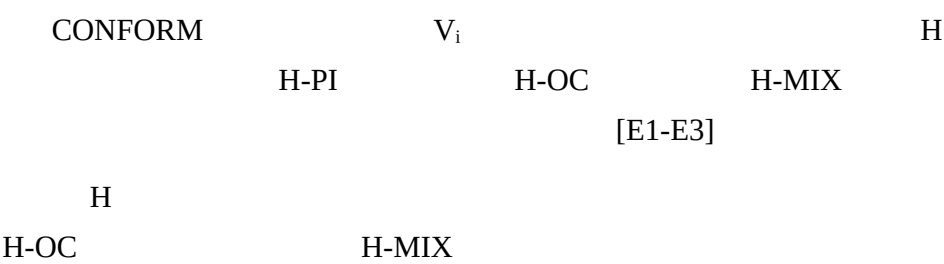
Registered Report

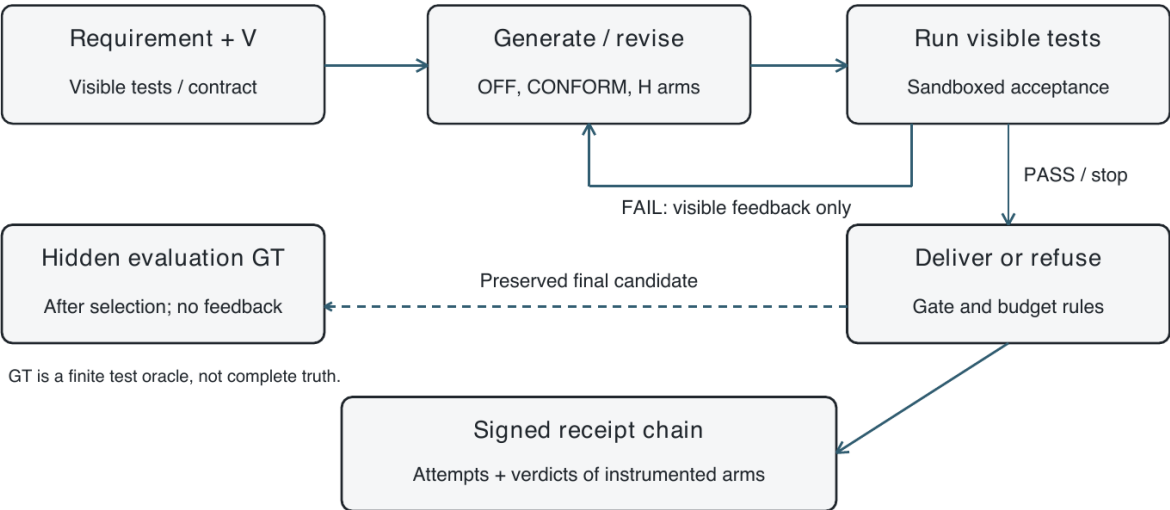
Barr	HumanEval	EvalPlus
LiveCodeBench [1-4]		
Self-consistency	Self-Debugging	Reflexion
Olausson [5-8]		
Kim	DRF	A-Trust [9-12]
PeerReview	CT [13-14]	
Körber	Nosek [15-16]	

3.1 □□□□□□□□□□



3.2 □□□□□□□□





1 GT

3.3 □□□□□

Vacant Ed25519
JSON
[E10]

194 R460R CONFORM H R529 CONFORM H-MIX OFF OFF5
9,841

3.4 □□□□□□□□

R453 R454
[E9]
Sybil

4.1 □□□□□□□□

2026 9 13 R460 R460R R529
CONFORM EQ5
[E1-E9]

R460 LCB v2 120

R529	LCB v3 medium 135	LCB v3 hard 54	HumanEval+ 156	MBPP+ 371
716	HumanEval+ 164	MBPP+ 378		
LCB v3	2024 8 10		[E5-E6 E11]	

4.2

2

OFF	1		
CONFORM	5		
OFF5	5		
EQ5	5		
H-PI	5		
H-OC	5		
H-MIX	5		
H	token	[E1-E3 E8]	
	gemma-4-12b-it-qat	Q4_0	persona persona
	0.7		[E1 E6]
H	32,000 token	900	token
	H-OC	CONFORM	OFF5
token	[E1]		
	H-MIX	CONFORM	EQ5
	CONFORM	OFF	

4.3

H1		H-PI	H-OC	H-MIX	OFF	H2		H-MIX
CONFORM		H3				EQ5		H4
[E1-E4 E8]								
R460	P-H4			CONFORM			R460R	H-MIX
P-R4				H1 H4			P-H P-R	P-X
	[E2-E4]							

4.4

task_id A B n b A B c A
B $\Delta = 100(b - c)/n$ infra_void
McNemar [17] $d = b + c$ $p = \min\{1, 2P[X \leq \min(b,c)]\}$
X Binomial($d, 0.5$) R460 R460R H OFF CONFORM
Holm R529 .05 [18]
Clopper-Pearson $\Delta = 100(2\pi - 1)d/n$ [19]

R529 McNemar b c
n = 600

4.5

R460 EFFECTIVE OFF +25 CONFORM +10
Holm token OFF5 CONFORM CONFORM
COSTLY_BUT_REAL +10 RULED_OUT
INCONCLUSIVE [E1]
RULED_OUT +10 R529
Holm [E3 E5]
R440P R460 R529
[E1 E5-E7]

4.6

rows.jsonl b/c p Holm
194 Ed25519
[E12]

5.1

R440P 179 31 147 22
82.68% 85.03% [E7]

5.2

CONFORM MBPP+ 371 +4.58 LCB v2
r447 +19.17 LCB v3 r461 +7.94
[E8]
R445 run
r447 R460 120

5.3

EQ5 3 p
[E8]
3 EQ5

	b	c	Δ pp	p
r446 MBPP+	280/371	265/371	24/9	+4.04 .0135
r448 MBPP+	286/371	273/371	21/8	+3.50 .0241
r449b LCB v2	85/120	75/120	15/5	+8.33 .0414
r449c LCB v3	157/189	149/189	13/5	+4.23 .0963

MBPP+ n [E8]
MBPP+ LCB v2 LCB v3 R446 40%
95% 20.49%
R460R CONFORM OFF5 EQ5 p
.05 EQ5

5.4

R453 Mac Linux 368 1,840
[E9]
R454 K3 273/273 K3 1,840
58/58 K3
[E9]

5.5 R460 □□□□□□

R460 120 OFF 70 CONFORM 85 OFF5 78 H-PI 91 H-OC 96 H-MIX 101
H-MIX CONFORM 22/6 +13.33 [4.22, 19.46] Holm p =
.0112 EFFECTIVE [E2]

R460 loader
H-MIX 14/120 CONFORM 29/120

+13.33

5.6 R460R □□□□□□

4 5 H-MIX CONFORM
116 [E4 E12]

4

	R460	r1	r2	r3	r4	r5
OFF	70/120 58.33%	69/120 57.50%	65/120 54.17%	69/120 57.50%	62/120 51.67%	61/119 51.26%
CONFORM	85/120 70.83%	86/120 71.67%	87/120 72.50%	90/120 75.00%	85/120 70.83%	83/117 70.94%
OFF5	78/120 65.00%	73/120 60.83%	74/120 61.67%	71/120 59.17%	80/120 66.67%	81/118 68.64%
H-PI	91/120 75.83%	92/120 76.67%	95/120 79.17%	91/120 75.83%	97/120 80.83%	85/120 70.83%
H-OC	96/120 80.00%	92/120 76.67%	88/120 73.33%	90/120 75.00%	94/120 78.33%	90/120 75.00%
H-MIX	101/120 84.17%	93/120 77.50%	92/120 76.67%	91/120 75.83%	88/120 73.33%	89/119 74.79%

r5 [E2 E4 E12]

5 H-MIX CONFORM

	n	b c	Δ pp	95%	Holm p	
R460	120	22/6	+13.33	[4.22, 19.46]	0.0112	EFFECTIVE
r1	120	15/8	+5.83	[-2.79, 12.89]	0.6301	INCONCLUSIVE
r2	120	17/12	+4.17	[-5.35, 12.80]	0.9165	INCONCLUSIVE
r3	120	12/11	+0.83	[-7.44, 8.89]	1.0000	RULED_OUT
r4	120	13/10	+2.50	[-5.94, 10.28]	0.6776	INCONCLUSIVE
r5	116	14/9	+4.31	[-4.54, 12.01]	0.9224	INCONCLUSIVE

RULED_OUT +10 pp [E2 E4 E12]

H-MIX CONFORM
R460 +13.33

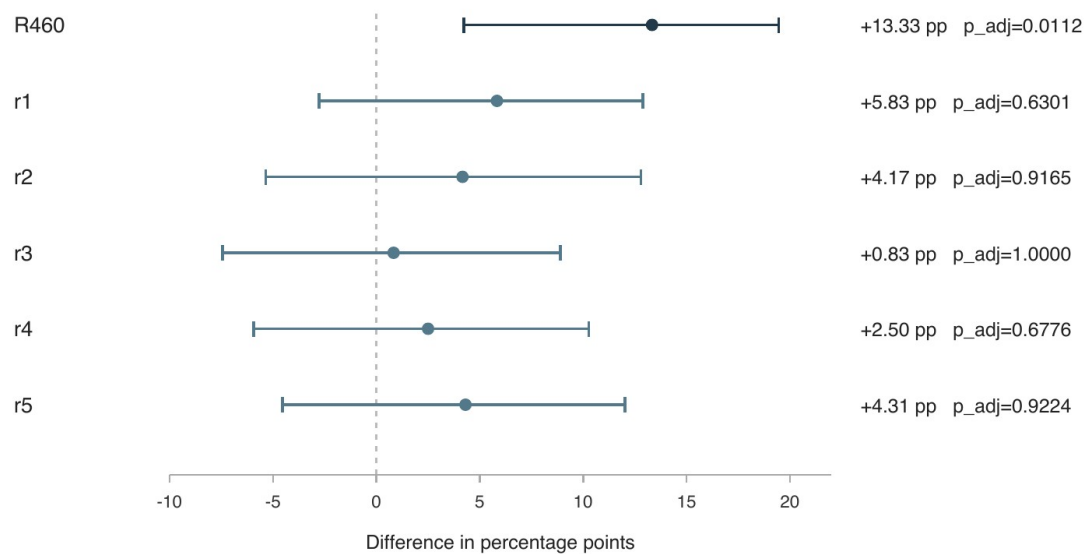
Holm

R460

H OFF Holm H CONFORM CONFORM
OFF +14.17 +18.33 +17.50 +19.17 +18.97

H-MIX minus CONFORM

Paired differences and unadjusted 95% conditional intervals



2 r1 r5 120 r5 116 Holm [E2 E4 E12]
H-MIX 20/120 14/120 19/120 21/120 19/119 CONFORM 26/120 26/120
23/120 25/120 25/117 P-R4 Δ 5/5
+10 0/5 Holm 0/5 token 3/5
+10 .43 .63
H-MIX CONFORM [+0.83, +7.50]
[E3-E4]

5.7 R529 □□□□□

716 6

[E6 E12]

6 H-MIX CONFORM

	n	OFF	CONFORM	H-MIX	b c	Δ pp
LCB v3 medium	135	115	125	126	6/5	+0.74
LCB v3 hard	54	38	41	43	5/3	+3.70

	n	OFF	CONFORM	H-MIX	b c	Δpp
HumanEval+	156	129	147	148	5/4	+0.64
MBPP+	371	277	295	299	15/11	+1.08
	716	559	608	616	31/23	+1.12

OFF 10 ⁻⁸	CONFORM [E6 E12]	H-MIX	n		Holm p	H-C .3409	H-O 2.02 ×
	H-MIX	CONFORM	31/23	+1.12	Holm	p = .3409	OFF
79/22	+7.96	p = 2.02 × 10 ⁻⁸	H-MIX	CONFORM	p	1.000	.727
1.000	.557	R529	INCONCLUSIVE				
R529							
	LCB medium	H-MIX 6/135	CONFORM 9/135	hard	7/54	11/54	HumanEval+
6/156	MBPP+	58/371	56/371				

reasoning

5.8 □□□□□□□□

R460R	R529	V/GT v2				R460R
95,090		57,248	37,842		39.8%	H
[E4 E6]						
v1	R460		v2			task_id
	V/GT		194	9,841		
R460R	120	R529	74	6,674	3,167	[E10 E12]

6.1 □□□□□□□□

		CONFORM	OFF
	H-MIX	CONFORM	
Olausson		[8]	
Self-Debugging	Reflexion	[6-7]	
H			

6.2 □ □ □ □ □ □ □ □

MIX

R529

CONFORM H-
[1 3]

token

6.3 □ □ □ □ □ □ □ □ □ □ □ □

[13-14]

Vacant

PeerReview CT

6.4 □ □ □ □ □ □ □ □

R460

6.5 □ □ □ □ □ □ □ □ □

7			
	[E1 E3 E5]	R440P	H4
		R529	
	Holm p [E12]		
	[E4 E6]		
	[E12]		
	[E9-E10]		
	[E11]		
	[E1-E6]		
	[16]		

[16]

7.1 □□□□□□□□□□

harness 12B Python persona

7.2 □□□□□□□□□□

LCB v3

7.3 □□□□□□□□□□

120

7.4 □□□□□□□□□□

n = 120 +10 R460R R529
r4 10.28 +10

7.5 □□□□□□□□□□

R529 reasoning reasoning R460R

7.6 □□□□□□□□□□

complete-case
token

7.7 □□□□□□□□□□

V/GT

7.8 □ □ □ □ □ □ □ □

7.9 □ □ □ □ □ □ □ □ □ □ □ □

G		RECORD_SPEC	
H-MIX		CONFORM	Holm
		194	9,841
		reasoning	
CONFORM	H-MIX		
Vacant		https://github.com/cosmopig/Vacant	
44be37fe52bac148ecd2835aa384c804c612ffea		A	
verify_evidence.py	verified_evidence.json	docs/paper_2026-09-14	

AI

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A

E

[E1] R460

DECISION_20260907_R460_HARNESS_PREREG.md

[E2] R460

DECISION_20260911_R460_FABLE_AUDIT_HARNESS.md

[E3]

DECISION_20260911_R460R_FIVE_REPLICATIONS_PREREG.md

[E4]

DECISION_20260912_R460R_FABLE_AUDIT_REPLICATIONS.md

[E5] R529

DECISION_20260911_R529_CROSS_BANK_PREREG.md

[E6] R529

DECISION_20260912_R529_FABLE_AUDIT_CROSS_BANK.md

[E7]

DECISION_20260903_R440P_CONFORMANCE_GATE.md

[E8] CONFORM EQ5

CONCLUSION_20260904_R445_CONFORM_SETTLEMENT.md

CONCLUSION_20260904_R446_EQUAL_BUDGET.md

DECISION_20260907_R449C_FABLE_AUDIT_UNRESOLVED.md

docs/VACANT_COMPLETE_2026-09-12.md

[E9]

DECISION_20260906_R453_FABLE_AUDIT_REAL_MULTIPARTY.md

DECISION_20260906_R454_FABLE_AUDIT_NAMED_DISSENT.md

DECISION_20260905_R449_PEEREXEC_ARCHITECTURE_AUDIT.md

[E10]

vacant/logbook.py

vacant/canonical.py

ops/gain/replay/r460r/receipts_verify.json

ops/gain/replay/r460r/receipts_verify_r4r5.json

ops/gain/replay/r529/receipts_verify.json

[E11]

runs/INDEX.md

vacant/codebench.py

docs/JOURNEY_2026-09-13.md

docs/HMIX_ARCHITECTURE_2026-09-11.md

[E12]

docs/paper_2026-09-14/verify_evidence.py

docs/paper_2026-09-14/verified_evidence.json

B

B1

R440P		[E7]
R460		[E1-E2]
V/GT v1	v2	[E2 E4]
R460R		[E4]
		[E6]
r5		[E4 E6]
R529	reasoning	[E6]

C

C1

	Self-Debugging	Reflexion [6-7]
	Olausson [8]	R460R R529
	Kim [9]	
	EQ5 CONFORM	[E8]
H-MIX		[E4 E6]
	PeerReview CT	[13-14 E12]
		[15]
	7	

D

verify_evidence.py	R460	R460R	R529		Holm
	R460R	R529	194	SHA-256	verified_evidence.json
					V/GT

A